# Ehsan Sharafian

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## PROFESSIONAL SUMMARY

Mechanical Engineering PhD candidate specializing in intelligent wearable systems for human movement analysis and digital health. Experienced in building end-to-end solutions that integrate IMU-based sensing, embedded hardware, smartphone applications, machine learning, and haptic feedback for gait analysis and rehabilitation. Skilled in Python, MATLAB, Android Studio/Java, and PyTorch, with six peer-reviewed journal publications.

## EDUCATION

**Ph.D. in Mechanical Engineering** | University of Maine, Orono, ME — Expected Dec 2027
GPA: 4.00/4.00 | Dissertation: Smart Real-Time Gait Training System

**M.S. in Mechanical Engineering** | Amirkabir University of Technology (Tehran Polytechnic) — Feb 2020
GPA: 3.65/4.00 | Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces

**B.S. in Mechanical Engineering** | University of Tehran — Jul 2017

## TECHNICAL SKILLS

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|---|---|
| **Programming & Data Analysis** | Python, Java, C/C++, NumPy, Pandas, scikit-learn, PyTorch, SPSS |
| **Mobile & Software** | MATLAB, Simulink, TwinCAT, Android Studio, Unity, Git |
| **Wearable & Embedded Systems** | IMU sensors, Arduino IDE, SparkFun modules, vibrotactile motors, PCB modules |
| **Biomechanics & Signal Processing** | Gait analysis, human activity recognition, sensor fusion, feature extraction, OpenSim |
| **Robotics** | Surgical robotics, parallel manipulators, inverse kinematics, dynamics, control systems |
| **CAD** | SolidWorks |

## RESEARCH & ENGINEERING EXPERIENCE

**Graduate Research Assistant** | Biorobotics and Biomechanics Lab, University of Maine — Sep 2023 – Present
Orono, ME

- Led the development of real-time wearable sensing systems and Android-based mobile applications for gait analysis, activity recognition, fall-risk assessment, and home-based rehabilitation.
- Built a smartphone-deployed gait assessment system using a single foot-mounted IMU to reconstruct foot-height trajectories and classify locomotion across level walking, stairs, and ramps, achieving **96.08% accuracy**, **<1.1 cm error**, and **112 ms latency**.
- Engineered a lightweight human activity recognition system using two IMUs, biomechanical feature extraction, and classifiers including SVM and NB, achieving **99.1% accuracy** for normal locomotion and **93.1% accuracy** for atypical stair-negotiation strategies in older adults.
- Architected a home-based haptic feedback gait-training system integrating IMUs, vibrotactile motors, custom Android applications, Arduino/SparkFun modules, and PCB circuitry for adults aged 65+.
- Demonstrated up to **30% improvement** in stride length and walking speed during a single-session haptic feedback study, with minimal cognitive demand.
- Led IRB-approved human-subject validation studies with **85+ participants** across gait analysis, activity recognition, and haptic feedback protocols.
- Developed real-time algorithms and data pipelines in Android Studio/Java, Python, and MATLAB, combining signal processing, sensor fusion, biomechanical modeling, and machine learning for mobile health applications.

**Mechanical and Control Systems Engineer** | Sina Robotics and Medical Innovators Co. Dec 2022 – Sep 2023

Tehran, Iran

- Developed PLC-based control logic for a telesurgical laparoscopic robotic platform using Beckhoff TwinCAT and IEC 61131-3 Structured Text, with MATLAB/Simulink validation of inverse kinematics and initialization workflows.
- Built a Python/Kivy graphical user interface on Raspberry Pi for surgeon-facing robot control, system guidance, warnings, and improved human-robot interaction during testing and demonstrations.
- Coordinated hands-on validation sessions with **6 experienced surgeons**, using training, troubleshooting, and usability feedback to refine surgeon-facing robotic system workflows.

**Laboratory Research Assistant** | New Technology Research Center, Amirkabir University of Technology Nov 2020 – Nov 2022

Tehran, Iran

- Designed and assembled robotics and mechatronics setups for student research projects, integrating hardware, CAD, MATLAB workflows, and technical documentation.
- Mentored 6 undergraduate students across 5 research teams, guiding equipment use, troubleshooting, documentation, and project execution.
- Coordinated technical preparation for the 9th International Conference on Robotics and Mechatronics, including lab demonstrations, equipment readiness, and presentation logistics.

## SELECTED HONORS

Flagship Fellowship Award, University of Maine Sep 2024 – Sep 2025

Graduate Student Government Award for conference proposal Mar 2025

## LEADERSHIP & SERVICE

**Graduate Senator**, Department of Mechanical Engineering, University of Maine Sep 2025 – Present

**President**, Iranian Graduate Student Association, University of Maine Sep 2024 – Apr 2026

## PUBLICATIONS

**6 peer-reviewed journal articles** in wearable sensing, biomechanics, rehabilitation engineering, and robotics.

- **Sensors** (2026) – Foot-height estimation and locomotion classification using a foot-mounted IMU and smartphone.
- **IEEE Access** (2025) – IMU-based activity recognition for older adults using minimal biomechanical features.
- **IEEE TNSRE** (2025) – Real-time haptic feedback effects on gait performance and cognitive load.
- **Journal of Biomechanics** (2025) – Thigh extension and leg coordination in young and older adult walkers.
- **International Journal of Robotics and Automation** (2023) – Robot-arm trajectory optimization for production-line applications.
- **Robotica** (2022) – Screw theory-based constraint wrench analysis for a Delta manipulator.